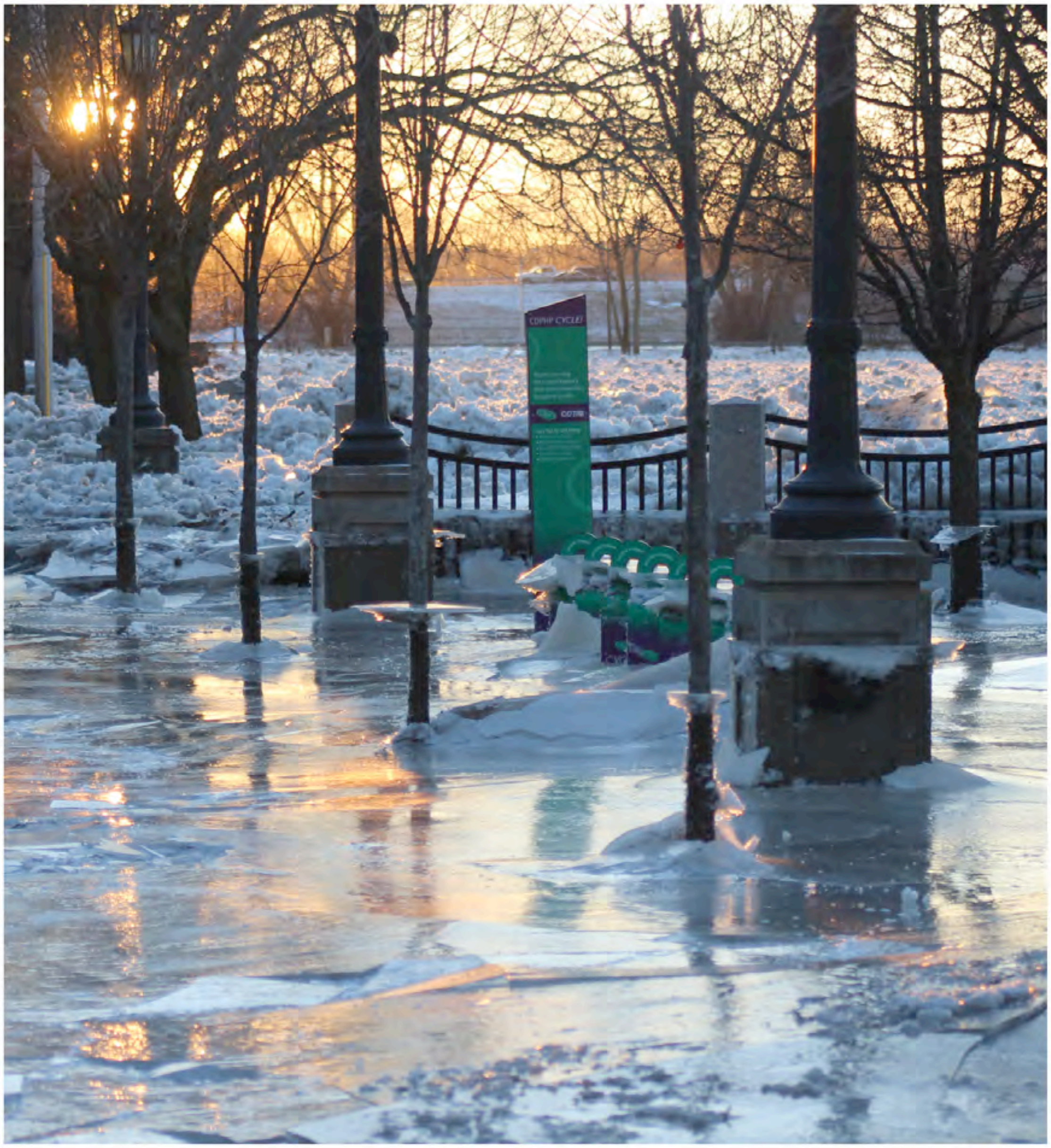
Mohawk Watershed Symposium 2018



Abstracts and Program
College Park Hall, Union College
Schenectady NY
23 March 2018

Flood forecasting of water discharge at Freeman's Bridge in Schenectady, New York

Scott R. Lakeram[1], Tom A. Plitnick[1], Daniella K. Chernoff[1], Antonios E. Marsellos[1], Katerina G. Tsakiri[2]

[1]Department of Geology, Environment and Sustainability, Hofstra University, Hempstead, NY
[2]Department of Information Systems and Supply Chain Management, College of Business Administration, Rider University, Lawrence, NJ

Introduction

The county of Schenectady encompasses 204.52 square miles and is located along the banks of the Mohawk River in Upstate New York (Census, 2010). With such a vast area citizens living in Schenectady are at the predisposition to the water levels of the Mohawk River. Due to the fluctuating water levels, flooding has become a common occurrence, particularly for individuals living along the banks of the river. These flooding events occur from a myriad of sources, many of which are derived from ice jams, inflicting damage to homes and local businesses. Ice jams occur during the winter when floes accumulate at the base of bridge piers, locks, and dam structures, inhibiting the movement of water downstream (Pariset et al. 1966). Research on flood forecasting and damage evaluation in this area has been conducted in the past (Foster et al. 2011, Tsakiri et al. 2014, and Pascucci et al. 2017). The recent ice jam that occurred on January 26th of 2018, raised the water levels of the Mohawk River substantially (USGS, 2018). In an attempt to warn residents of possible future flooding events, daily climate and groundwater data from the past six years were statistically analyzed in order to achieve long-term forecasting.

Methodology

In order to evaluate issues of flooding along the city's river banks, atmospheric and hydrological data were obtained from weather stations situated near Schenectady, NY (Water Discharge Station Id 01354500 and Station Id 01354330, USGS Well 425048073472501, USGS Well 424859073585501, Albany Airport Weather Station GHCND:USW00014735, and Pressure Data Station WBAN 04741). Using the weather stations, well stations and bridge location as pins, maps of the studied area were created utilizing a digital elevation model (DEM) with Light Detection And Ranging (LiDAR) data (LiDAR-DEM) in Global Mapper (Figure 1). Once the data were obtained, variables were inserted into KNIME, an open source data analytics and data mining software, reporting and integration platform. Here, the raw data was mined from their original url sources. Hourly data were converted to daily data by extracting the maximum value of the day and missing data were replaced with previous daily values. A Log transformation was placed on the water discharge data to reduce the difference between extreme values and minimum values for that day. A periodicity of thirty-one days was selected, a Kolmogorov-Zurbenko filter was applied in order to determine long-term values and a linear correlation was conducted in order to find any similarities between the variables (Yang et al. 2010). The filtered data were transferred into Statistical Package for the Social Sciences (SPSS) where a linear regression was performed. The long-term and raw data were analysed for differences in the regression number. The unstandardized values were extracted and plotted with the long-term discharge values.

Results - Interpretation

A detailed map of the studied area was produced using Global Mapper. The LiDAR bare earth model was generated in Global Mapper to explore the landscape and confirm that all stations are located within an appropriate distance and no physical barriers were present, such as ridges. All studied points were placed on the map as pins and labeled. The programs KNIME and SPSS were used to analyze data collected from the weather and groundwater stations. The linear regression yielded a 95% of coefficient of determination between the long-term component of the water discharge and the independent variables. Thus, 95% of the variation in the long-term of the water discharge can be explained by the atmospheric and hydrologic variables such as: the ground water depth upstream, the difference in groundwater depth (between upstream and downstream), the maximum temperature, the minimum temperature, the precipitation, the snowfall, the snow depth and the average wind speed. The P-values of the coefficients in the linear regression model are all less than 0.05 indicating that all the variables were statistically significant.

To verify the assumptions in the linear regression model, we check the plots of the residuals. The histogram of the regression standardized residuals follows the normal distribution verifying the normality of the residuals (Figure 2). The scatter plot of the residuals divided by the standard deviation was completed to determine if any outliers were present in the data. The plotted data produced a plot with no patterns and extreme points (Figure 2).

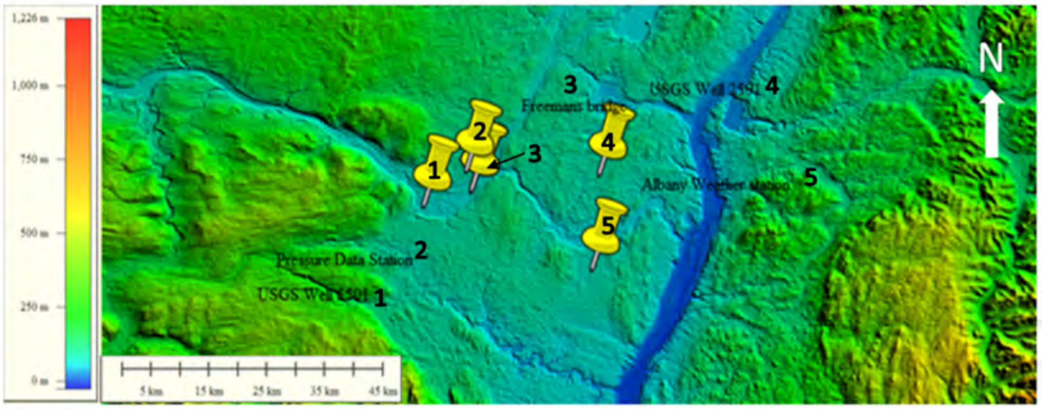


Figure 1: A LiDAR bare earth model of the greater Schenectady area showing the Mohawk River (flowing from the west to the east) depositing in the Hudson River. The vertical scale shows vertical elevation with no exaggeration (in Meters) and the horizontal scale is in Kilometers. Yellow pins located on the map indicate weather and groundwater station structures where data was taken from. The yellow pins located on the map correspond with a numerical value to indicate locations (USGS Well 425048073475501 (1), Pressure Data Station WBAN 04741 (2), Freeman's Bridge (3), USGS well 424859073582501 (4), Albany International Airport Weather Station GHCND:USW00014735 (5).

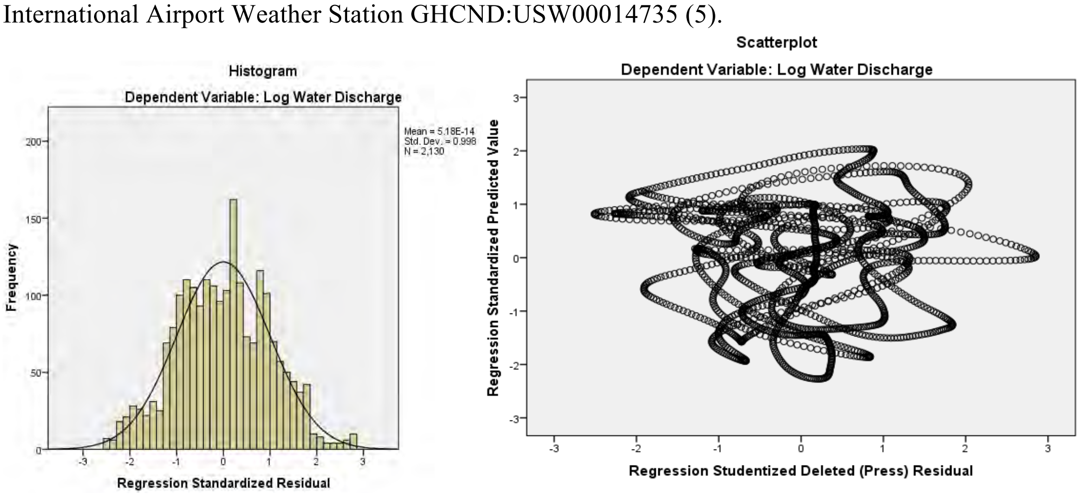


Figure 2: On the left side, the histogram of the standardized residual derived from the linear regression model (N=2,130). On the right side, a scatter plot of the studentized residual derived by the regression model versus the predicted values for the dependent variable (N=2,130).

Discussion

Schenectady County, which is located in Upstate New York, is highly susceptible to flooding events that occur along the Mohawk River. These events are caused by high amounts of rainfall and primarily by ice jams. Using KNIME and SPSS the data were analysed in order to predict flooding events caused by ice jams near Freeman's Bridge.

The plotted predicted discharge values closely follow the trend of the long-term water discharge, indicating the accuracy of the model. From the graph it was determined that two flooding events occurring in June 2013 and July 2017 along with two ice jams occurring in January 2014 and January 2018. Even though there are distinctive flooding events, series of consecutive flooding events have happened and aggregations of flooding events were sometimes clumped together and were displayed as a single exponential trend.

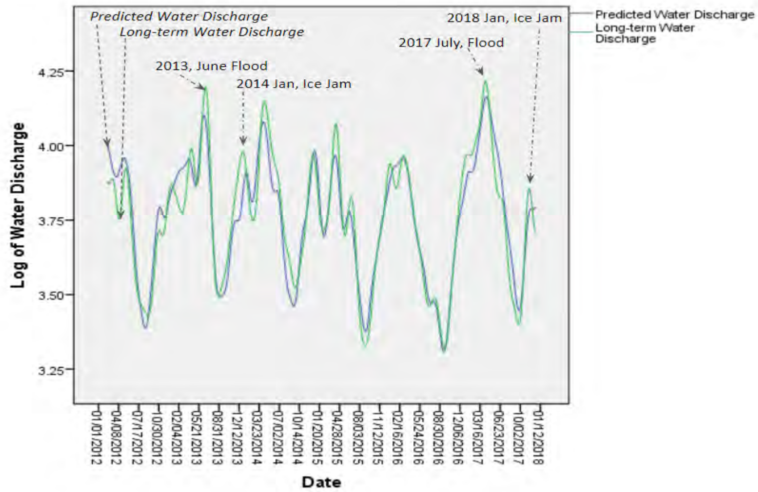


Figure 3**:** Time Series plot of the unstandardized predicted values overlain with the long-term of the water discharge from January of 2012 to February of 2018. Four major events are showing on the graph (June 2013, January 2014, July 2017, and January 2018) indicating an ice jam and/or flood event.

Conclusion

The accuracy of this method could prove to be a reliable source for forecasting future ice jams and flooding events. This methodology can be applied to forecasting floods in Schenectady County, increasing the preparedness of the public and reducing the risk of damage to local infrastructure. Anticipating flood events in advance may provide adequate time for public authorities to immediately respond to flood calls and minimize injuries that can occur to the public.

References:

Census. (2010) Schenectady County, New York; UNITED STATES. *www.census.gov*, www.census.gov/quickfacts/fact/table/schenectadycountynewyork,US/PST045216.

Foster, J.A., Marsellos, A.E., Garver, J.I. (2011) Predicting trigger level for ice jam flooding of the lower Mohawk River using LiDAR and GIS. Proceedings from the Mohawk Symposium 2011, in Schenectady, New York, p.13-15.

United States Geological Survey, Mohawk River Ice Jam Monitoring." *New York Water Science Center*, USGS, Updates, 27 Jan. 2018, ny.water.usgs.gov/flood/MohawkIce/.

Pariset, Ernest, Hausser, René, Gagnon, André. (1966) Formation of Ice Covers and Ice Jams in Rivers. Journal of the Hydraulics Division, Vol. 92, Issue 6, Pg. 1-24

Pascucci, T., Chernoff, D., Lakeram, S. & Marsellos, A. (2017) Statistical analysis of damage to local businesses due to flooding events along the Mohawk River valley in Amsterdam, New York. Proceedings from the Mohawk Watershed Symposium 2017, p. 50-52 ISBN: 978-1-938868-12-8

Tsakiri, K.G., Marsellos, A.E., Zurbenko, I.G.,(2014) An efficient prediction model for water discharge in Schoharie Creek, NY. Vol. 2014 Article ID 284137, Journal of Climatology. doi:10.1155/2014/284137.

Yang, Wei, Zurbenko, Igor. (2010) Kolmogorov–Zurbenko filters. Wiley Interdisciplinary Reviews: Computational Statistics 2, no. 3: 340-351.

Poster Presentation